This section provides learners with a knowledge and understanding of the use of microscopy in the development of the cell theory, and the importance of microscopy in investigating the structure of eukaryotic and prokaryotic cells. Learners will also gain an insight into some of the advanced microscopic techniques used by cell biologists.

The structure of animal cells is studied in the context of the histology of mammalian blood as revealed by light microscopy and is contrasted with the structure of typical plant cells.

An understanding of the importance of enzyme-catalysed reactions is illustrated by the biochemical processes that prevent excessive blood loss, and learners investigate how first-aid and medical intervention can assist the body's natural mechanisms for preventing blood loss.

In addition to enzymes, this section covers the structures and functions of other biologically important molecules, including DNA, natural polymers, and the different mechanisms for transporting molecules into and out of living cells. Learners gain an appreciation of the importance of water as the essential medium in which transport and exchange takes place.

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.

2.1.1 Cells and microscopy

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)** the importance of microscopy in the development of the cell theory and the investigation of cell structure	To include the use of the light microscope, transmission and scanning electron microscopes and recent developments such as the confocal scanning microscope.
	(ii) the preparation of blood smears (films) for use in light microscopy	To include how blood smears are made and the interpretation of stained material. Practical work to be carried out in accordance with current CLEAPSS® guidelines. Also see Section 5f. **PAG1** HSW4, HSW6
(b)	the procedure for differential staining	To include the use of Leishman's stain to identify leucocytes in blood smears. HSW4, HSW6

(c)	**(i)** the structure of animal cells as illustrated by a range of blood cells and components as revealed by the light microscope	To include red blood cells (erythrocytes), platelets, neutrophils, lymphocytes and monocytes as specialised cells with particular functions related to their structures.
	(ii) the observation, drawing and annotation of cells in a blood smear as observed using the light microscope	**PAG1**
(d)	the linear dimension of cells and the use and manipulation of the magnification formula $\text{magnification} = \dfrac{\text{image size}}{\text{actual size (of object)}}$	*M1.8* **PAG1**
(e)	practical investigations using a haemocytometer to determine cell counts	To determine the mean numbers of erythrocytes and convert to a concentration (to include details of dilutions and calculations). *M0.1, M0.2, M0.3, M1.5, M1.10, M4.1* **PAG1** HSW3, HSW4, HSW5, HSW6
(f)	the principles and use of flow cytometry in blood analysis	To include the use of fluorescent labels. Details of the use of different lasers is not required. HSW3, HSW8
(g)	the ultrastructure of a typical eukaryotic animal cell, such as a leucocyte, as revealed by an electron microscope	To include the structure and function of the following: cell surface membrane, Golgi apparatus, rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER), ribosomes, lysosomes, vesicles, mitochondria, cytoskeleton, centrioles. nucleus and nucleolus. HSW8 *M1.8*
(h)	**(i)** the ultrastructure of a typical eukaryotic plant cell such as a palisade mesophyll cell and a prokaryotic cell, as revealed by an electron microscope	To include chloroplasts, large vacuole and the cell wall in plant cells, circular DNA, plasmids, mesosome, pili and flagella in prokaryotic cells. HSW8
	(ii) the similarities and differences between the structure of eukaryotic plant and animal cells, and between eukaryotic and prokaryotic cells	

(i)	practical investigations using a graticule and stage micrometer to calculate and measure linear dimensions of cells	To include the calibration of an eyepiece graticule using a stage micrometer, calculating the area of the field of view and measuring the sizes of organs, tissues, cells and organelles and calculating their magnification. *M0.1, M0.2, M1.1, M1.2, M1.8, M2.2, M4.1* **PAG1** HSW3, HSW4, HSW8
(j)	how the plasma membrane is composed of modified lipids and how the structure of triglycerides and phospholipids is related to their functions	To include reference to fatty acids, glycerol, phosphate groups, ester bonds and hydrophobic/hydrophilic properties.
(k)	the fluid mosaic model of the typical plasma membrane	To include the location and function of phospholipids, intrinsic proteins, extrinsic proteins, cholesterol, glycolipids and glycoproteins.
(l)	the movement of molecules across plasma membranes	To include diffusion and facilitated diffusion as passive methods of transport across membranes **AND** active transport, endocytosis and exocytosis as processes requiring ATP as an immediate source of energy. HSW8
(m)	practical investigation(s) into factors affecting diffusion rates in cells	To include the use of model cells and tissues such as beetroot. *M0.1, M0.2, M1.1, M1.2, M3.1, M3.2, M3.3, M3.5, M3.6* **PAG8** HSW1, HSW2, HSW3, HSW4, HSW5, HSW6
(n)	the roles of membranes within and at the surface of cells	
(o)	the interrelationship between the organelles involved in the production and secretion of proteins.	To include the role of the cytoskeleton and motor proteins, the nucleus, ribosomes, RER, Golgi and vesicles (no details of transcription and translation are required at this stage). HSW8